Taxation and Efficiency: In-Class Questions

EC313 Public Economics: Taxation, Fall 2026

These questions are written in the style of the short-answer section of the midterm, where each part is worth 5 marks. We work through the answers in class, so solutions are not posted.

Question 1

For each of the following, state whether it is true or false, and provide an explanation (all marks come from the explanation).

a) If a tax leaves the quantity of the taxed good unchanged, the tax has no excess burden.

b) A subsidy lowers the price consumers pay, so unlike a tax it cannot create an excess burden.

Question 2

A province introduces a 20% tax on food delivery orders. Before the tax, households spend $2 billion a year on delivery orders. The compensated price elasticity of demand is 1.2 (in absolute value), and supply is perfectly elastic.

a) Calculate the excess burden of the tax, the revenue it raises, and the excess burden per dollar of revenue.

b) In the short run, restaurants and couriers cannot quickly change how many orders they can handle, so supply slopes upward. Is the short-run excess burden larger or smaller than your answer in (a)? Use a graph to support your answer.

Question 3

For each of the following, state whether it is true or false, and provide an explanation (all marks come from the explanation).

a) A province already taxes movie theatre tickets at 15%. A new 15% tax on streaming subscriptions, a substitute, must increase the total excess burden of the province's taxes, because it adds a second distortion.

b) A tax set so high that people stop buying the good entirely has no excess burden, because it raises no revenue.